

Coursework: Analyzing the broader impact of AI (5 November 2025)

Description of dataset and baseline: The chosen model is the one explained in “CoastSat: A Google Earth Engine-enabled Python toolkit to extract shorelines from publicly available satellite imagery” [1]. CoastSat is an open-source software designed to monitor global coastal shorelines. It employs an automated extraction pipeline that, in the first step, downloads satellite images from the Google Earth Engine (GEE) API and performs preprocessing. Then, use a machine learning model that works as a two-step process: first, a multilayer perceptron that classifies pixels into four components, and second, a sub-pixel resolution border segmentation using a modified normalized difference water index and a marching square algorithm to estimate the shoreline position. The software also offers tidal correction of the output using slope and water level data, which can be obtained from global models or in-person measurements. Users can estimate concrete historical data of the coastal shoreline from 30+ years, presenting an opportunity for more informed coastal management [1].

The CoastSat paper provides documentation regarding its capabilities and workflow, including guided examples and step-by-step instructions on its GitHub [2]. However, it features notable gaps when it comes to elaborating on its operational limitations and comparative benchmarking against alternative models. The comparison focuses on one location, where CoastSat obtained a Root Mean Square Error (RMSE) of 7.2 m, compared to the study by Luijendijk et al. [3], which achieved an RMSE of 13.7 [1]. It does not formally compare both methodologies and model architectures. The sources of error are explained qualitatively, with a focus on the utilization of the tools, rather than a quantitative analysis of the causes.

Connection to SDGs: Coastal nations face the challenge of managing an ecosystem that is undergoing rapid change due to the rise of sea level, driven by the melting of polar ice and glaciers [4]. This risk can be exacerbated by natural disasters, such as floods or tsunamis, which disrupt the coastal system [5]. By providing crucial data for shoreline monitoring and analysis, CoastSat directly supports the United Nations' SDG (Sustainable Development Goal) 13, Climate Action. This technology democratizes access to low-cost and long-term coastal information, which is necessary to identify the rate of deterioration of the coastal system and ensure a local understanding of the vulnerabilities in specific coastal areas, supporting target 13.1 (strengthening resilience and adaptive capacities). Moreover, CoastSat informs effective target disaster preparedness, response, and recovery measures, especially in countries with limited resources (target 13.b)[6].

Most of the coastal territories have developed significant human infrastructure over the years, ranging from urban development to commercial ports, that can be severely affected by the impact of climate change [7]. Therefore, understanding shoreline evolution is a critical factor in developing responsible management of these territories, ensuring a sustainable preservation of these assets and communities [3]. Accordingly, CoastSat can make a direct contribution to achieving the UN's SDG 11 (Sustainable cities and communities) and help in meeting targets 11.5 (reducing the impact of disasters), 11.3 (enhancing sustainable urbanization), and 11.b (implementing disaster risk reduction plans)[6].

Despite the benefits, machine learning software has inherited several risks that may inhibit progress toward SDGs [8], [9]. The development of CoastSat-like systems, which rely on data-intensive processing, contributes to the environmental footprint of the technology sector [10]. The computing resources required for the storage and analysis of satellite imagery result in high electricity demands—creating direct negative externalities that challenge the achievement of SDG target 13.2 (integrating climate mitigation into national policies). Control over processing capacities resides with large private corporations, introducing risks of amplifying the structural inequalities between global technology providers and local communities, directly impacting SDG target 11.3 (inclusive urbanization and participatory planning). Furthermore, CoastSat's capacity to generalize is constrained by the finite scope of its training datasets; this limitation can lead to errors when applied to ecologically or geographically distinct regions, posing risks to reliable shoreline mapping and sustainable coastal management practices critical for SDG target 11.5 (reducing the adverse impacts of disasters)[6].

Impact on social sustainability: Are individuals and communities involved in making decisions about the activity, and is the decision-making process fair and democratic? Coastal management involves a diverse range of stakeholders, including local communities, researchers, policymakers, the public, and the industrial sector [11]. In this context, CoastSat seeks to provide a global and accessible methodology to obtain key information to support decision-making [1], [12]. This tool bridges the gap between groups with limited research resources and those with high investment capacities. Furthermore, the creation of long-term historical shoreline data also provides a more complete picture [13]. This long-term perspective is often part of the local cultural heritage in the form of stories or rituals, and without the quantification that CoastSat generates, this knowledge is easily discarded. Despite these positive effects, the quantification of a complex system tends to simplify a multidimensional phenomenon into a simpler model, creating a risk of misleading interpretations because of partial knowledge [10]. There is a risk that deceitful or misinformed actors may dismiss relevant factors of the coastal ecosystem, such as geological, morphological, and biotic components, thereby forcing decisions that have a negative impact on communities and the ecosystem.

Is there an external investigation of error sources? What kind of errors could you envision? The original paper doesn't have an extended comparison with other models and different coastal conditions, but because CoastSat is open source, any actor can evaluate the errors and adjust its capacities to their own local conditions [1], [13]. There are two developments I think are relevant to this question, the first is a benchmark published in a paper during 2023 that creates a formal framework to compare the effectiveness of different models for shoreline estimation [14]. Second, the evolution of the tools guideline to incorporate a specific segment to train the core classification model for local coastal conditions [15]. Furthermore, the sources of errors mentioned by the author are horizontal inaccuracy of georeferencing, cloud masking for large tidal cycles because of the loss in temporal resolution, slope conditions, and wave run-up characteristics [1], [16]. These last two sources are dependent on the water level of a specific temporal and special location; therefore, they are the main limitation for a good generalization, as it depends on the specific morphology and climate conditions of the different areas of each beach [12], [14].

Impact on environmental sustainability: Is the application resource-intensive? What about the data collection? The CoastSat software is designed to be efficient and accessible. The original paper states that 30 years of biweekly satellite images can be processed in approximately 2 hours using a common laptop [1]; hence, it doesn't generate a big electrical consumption. The guideline also specifies that the architecture is a Multilayer Perceptron (Neural Network) with 2 layers, making it a small model with a low training consumption. Data collection is a completely different scene. The usage of satellite images generates two main sources of pollution. First, the amount of electricity used by the computer infrastructure of GEE to store and process the big data pipeline of the different public satellite images. The global information and communication sector is projected to reach 20–30% of worldwide electricity demand by 2030 [17], [18]. Second, the global race for deploying satellites into the low Earth orbit (LEO) for research or commercial use has created a serious risk of space pollution, as this practice is hard to regulate and supervise. Based on persistent deployment rates, the LEO satellite population is projected to surpass 100,000 within the next decade, amplifying risk from both direct collisions and the accumulation of debris in low Earth orbit [19], [20].

Does the application protect us against risks related to climate instability and disasters? With CoastSat output, coastal societies can estimate with spatial and temporal detail the probability of risk related to coastal erosion or the effects of natural disasters. Using these insights, local authorities can implement updated zone regulations for safer land use that ensure an efficient response to coastal events, reducing the exposure of inhabitants and assets [11]. Also, this information can improve disaster prevention plans, as better predefined resettlement zones or more efficient evacuation routes. Additionally, these communities can invest in preventing infrastructure such as seawalls, groynes, and dune restorations, using CoastSat to identify the higher-risk locations to implement these solutions. Hence, this tool is a key enabler to implement effective measures to protect coastal nations from the effects of climate and disasters [1].

Impact on economic sustainability: What could be the economic impact of the application? In the short term, CoastSat could generate a negative economic impact because it can identify areas that are at high risk of being affected by natural disasters or erosion, highlighting the urgency to relocate communities or assets. This action brings high social, cultural, and economic costs, as people and industry leave behind most of their properties and rebuild from scratch [7]. Moreover, it could also produce a more restricted land use regulation, which can be an impediment to the commercial use of some territories. Additionally, preventive infrastructure could be necessary, such as seawalls and groynes, producing a high-cost implementation [11]. In the long run, with clear land and coast planning, it will ensure a more sustainable environment with lower risk, creating a better context for investment and fostering economic growth in the different localities [7], [21].

Is the model open-source? How could that impact our economy, especially if the solution can be deployed and reused in a scalable fashion to other problems? This dimension builds upon the previous point, as CoastSat is an open-source and easy-to-use software [1] that democratizes access to environmental information, empowering people to drive sustainable development and counterbalance market forces. If all the different groups that participate in the management of coastal territory have access to this information, it makes the behaviour of coastal ecosystems an obligatory factor to consider before any coastal project. It creates a door to reframe how coastal communities are evolving, probably slowing down the economic and urban development to organize and create a sustainable plan, but with the reassurance that there will be better risk management that fosters growth and investment in the coastal zone of the future. Therefore, if CoastSat development inspires other open-source AI solutions with similar goals, it can increase the level and the number of factors to consider in the development of a territory, making sure that it is going to be responsible and sustainable.

Other sustainability factors: Does this tool help manage resources that face growing scarcity in the future? How? This question is relevant because current development brings the challenge of higher demand for limited resources, a situation that nowadays creates difficulties on various continents, such as the energy crisis in Europe during winter, and water stress in India and Asia. CoastSat can help with one critical resource, potable water. There are various sources that depict a future world where humans would be fighting for this resource, but CoastSat enables knowing the areas of low risk and tidal

stability over time, facilitating building safe infrastructure for desalination [1], [22]. Even though there are other challenges to make desalination plants more efficient, such as energy consumption and wastewater management, these tools support a future where desalination plants can be a big part of our potable water supply.

At what level does this model consider the interactions or complexity of the ecosystem it tries to emulate? It is essential to have clarity on three modelling components that help to capture a system's complexity. The first is the pattern to predict; in this case, it is shoreline position for a specific time and location. This number represents limited information of the complex tidal system, leaving out morphological, geological, fluid dynamics, and biological components [12], [23]. The second component is the representation of features and the structure of the model. For this pipeline, the features are pixels of a satellite image that go through a neural network. This structure is rich in complexity but doesn't explicitly accentuate any semantic feature that can bring a theoretical structure. Similarly, the model doesn't incorporate previous knowledge about the process that is going to be predicted [4]. The third dimension is the nature of the training data set. In this case, we already talked about how the limited size of the training data doesn't cover all morphological conditions [1], negatively affecting the representation of different types of coastal environments. Therefore, the same characteristic that makes this model accessible and low-cost also makes it limited to a very specific indicator, imposing the necessity to complement it with other sources that can enrich the representation of the coastal ecosystem, such as morphological studies or geological records, diminishing the risk of making decisions with partial knowledge [24].

Can this model be used by the established hegemony to increase or preserve the power structure for their benefit? The real world is shaped, in part, by power dynamics that frame the actions available for different groups [25], [26]. In this case, the short-term measures for mitigation against the identification of high-risk coastal zones can disproportionately benefit specific groups if local authorities don't establish an equitable regulatory framework [17]. Groups with high investment capacity, for example, can afford to build preventive infrastructures, while low-income groups cannot [7]. This situation creates a scenario where the latter are displaced from their homes because of the imminent risk, and the former can take advantage. The wealthier actors can argue that they have the capacity to maintain and make use of these territories. In the academic literature, this phenomenon is called "climate gentrification" and could lead to extensive coastal areas being secured in the hands of a few, thereby displacing the communities that CoastSat was meant to help [27].

Intersection between SDGs: Climate Action (13) is indivisible (+3) to Clean Water and Sanitation goal (6)[28]. CoastSat delivers key information to identify high-risk coastal zones, allowing them to create resilience and adaptive capacity to climate-related hazards and natural disasters (13.1), especially in marginalized communities or small islands (13.b). These latter areas share the condition that most of their potable water supplies are susceptible to natural disasters, like tsunamis and floods. Hence, these targets of Goal 13 are a clear indivisible condition to achieve universal and equitable access to safe and affordable drinking water for all by 2030 (6.1). Additionally, the positive impact of CoastSat on the accomplishment of the targets for climate change through accessible information also supports and strengthens the participation of local communities in improving water and sanitation management (6.b).

Sustainable cities and communities (11) enable (+1) Decent work and economic growth (8)[28]. One of the direct impacts of the usage of CoastSat software is the implementation of integrated policies for adaptation to climate change and natural disasters in coastal areas through a better understanding of the risks in this ecosystem (11.b) and consequently diminishing the negative effect of water-related disasters on assets or communities (11.5). As a result, a more trustworthy and sustainable context creates the conditions for higher and more diverse productive activities, fostering local startups and businesses (8.3), such as sustainable tourism, which is a common industry in coastal areas (8.9).

Climate Action (13) constraint (-1) Reduced Inequality (10)[28]. The possible action taken to mitigate the effects of climate change (13.2 and 13.b) can directly affect marginalized and lower-income communities, as the short-term measures require high economic investment, and the option of resettlement has a greater impact, as this group of inhabitants depends strongly on the ecosystem as their source of livelihood. Therefore, it is a possible limitation to ensure equal opportunities and reduce inequality of outcome, if the climate actions are not designed thoughtfully, considering these vulnerabilities (10.6).

Speculative solutions: It is essential to have more labelled datasets for different morphological and geological systems. This labelling process is expensive, and some communities don't have the resources or the knowledge available to generate this information and the tide measure that supports the validation process. Therefore, the core relevance is to amplify and incorporate these territories as zones of interest. Adding semantic information to the dataset can help categorize different types of coastal zones, creating a better baseline to identify limitations or areas for improvement.

Explore more advanced models. Diffusion models have accelerated the development of representing complex systems. A perfect example is GenCast, a diffusion-based AI model from Google DeepMind that generates probabilistic weather forecasts [4], [29]. This type of model can create a more robust and generalizable output as it generates a probability

distribution that represents the probabilistic future of chaotic physical systems. Conversely, other types of models that could bring benefits in deeper representation are physics-informed neural networks. These models are neural networks that encode the law of physics through differential equations into their loss function, ensuring the solution adjusts to the pattern in the data and follows physical consistency. This methodology can incorporate previous knowledge of the physical rules that govern the behaviour of waves and tides [4], [30]. Both previous solutions make the model structure a lot more complex, going in the opposite direction of the author's vision; hence, a third possible solution is applying Bayesian models to incorporate previous knowledge and a better quantification of the uncertainty, but maintaining a simple architecture [10].

When using CoastSat, the regulation should ensure equality of outcome and democratic participation, to mitigate the possible negative impact that climate change actions can have on marginal communities because of economic inequality. Additionally, there should be a public and transparent framework that allows us to unify the local use and output of CoastSat, because of the flexibility of the open-source community. This could be created through technical institutions that educate in the correct use of this application and enrich the analysis with complementary information that can promote a more complete understanding of the coastal system, such as water level and slope, morphological detail, and other measures that have a higher cost of acquisition, like GPS or Lidar data. Moreover, the satellite industry should be regulated with a vision that fosters collaboration for the efficient usage of space resources, achieving a greater efficacy of territorial mapping with fewer externalities.

Under ideal conditions, I would have unbiased labeled training data that samples every archetype of coastal system with complementary data of morphological and geological conditions, and semantic data that allow me to identify different coastal components. With this, I would be able to introduce previous knowledge to a probabilistic model that will allow me a better generalization, interpretability, and quantification of uncertainty, without a considerable incrementation of the model complexity. Also, the efficiency of data collection is key, having a good regulation for the satellite industry and technology infrastructure industry, which ensures an efficient and trustworthy data source. At last, a clear framework for using AI models for supporting policy creation that ensures a transparent and democratic process.

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